CLASSIFICATION [redacted]

CENTRAL INTELLIGENCE AGENCY

INFORMATION REPORT

REPORT NO. SO 57499x

CD NO. 000008

COUNTRY USSR (Uzbek SSR)

SUBJECT Light Phenomena East of Tashkent

PLACE ACQUIRED [redacted]

DATE OF INFO. May - September 19[illegible]

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SUPPLEMENT TO REPORT NO. SO-49149x

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[redacted]

THIS IS UNEVALUATED INFORMATION

SOURCE [redacted] second interrogation. **

1. During the period from May to September 1947 three light phenomena happening at intervals of about 15 minutes were seen almost every night between 9 and 10 p.m. local time. The phenomena were watched from the PW camp in Pakhta Aral, about 50 km southwest of Tashkent (41°18'N/69°15'E), Uzbek S.S.R., in an east-southeasterly direction.

2. A darkred ball of fire was seen first; after about six seconds it reached the apex of a long-drawn out trajectory. During this time the ball had developed a trail of fire. Its color, which was bright red at the apex point, changed in color from pale green to white. Smoke trails, noises, or detonations were not noticed.

3. The distance of the trajectory from the point of observation was estimated at 60 to 80 km. * The trajectory ran approximately from southwest to northeast; its length to the apex, which was from 5,000 to 12,000 meters high, was estimated at about 80 km. At the apex the ball of fire seemed about one-fifth the diameter of the full moon.

* Field Comment. The measurements stated with respect to the trajectory must be received with reserve. It must be considered that neither the discharge nor the noise developed by the projectile were heard. For trajectory of phenomena, see annex.

** Washington Comment: This is a reinterrogation of the informant of SO-49149x.

1 Annex: One sketch showing trajectory.

APPROVED FOR RELEASE
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[redacted]

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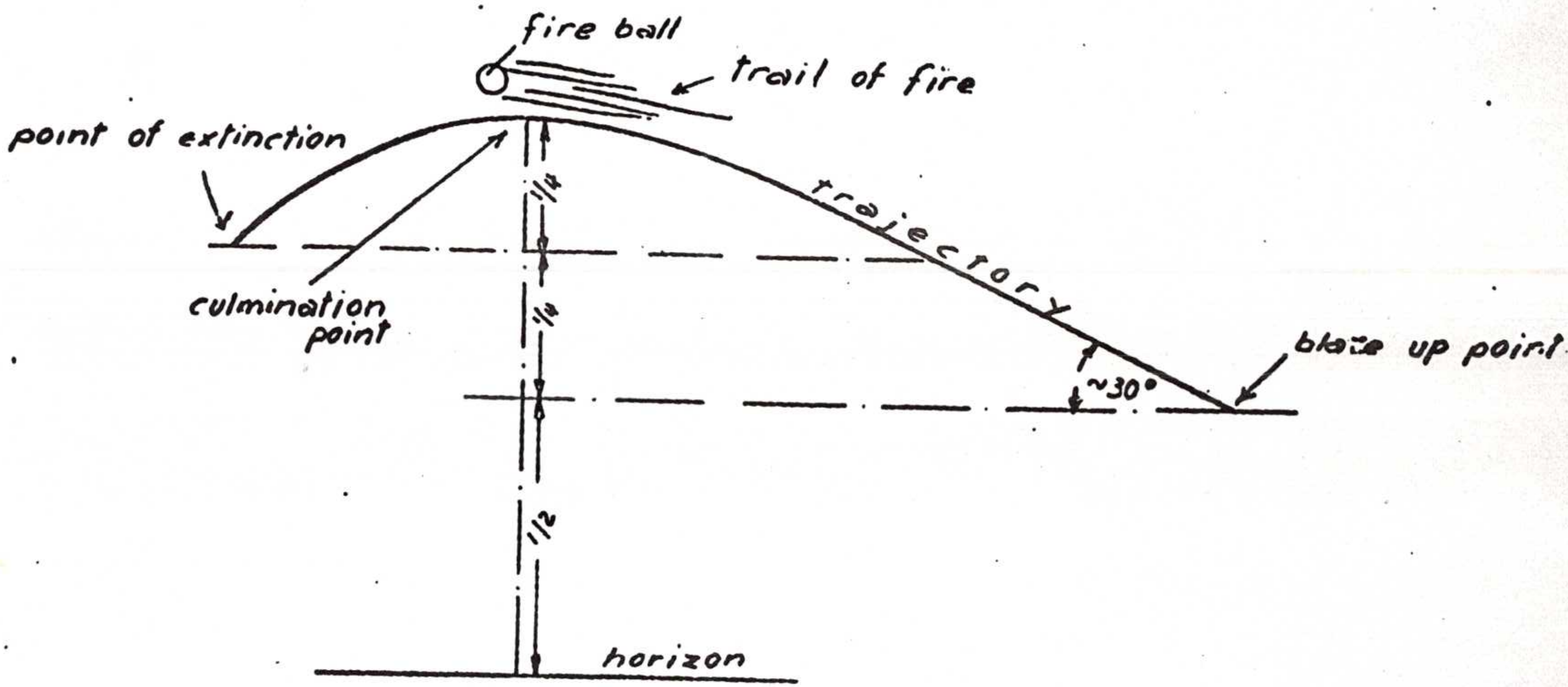
SO-57499x
Annex

000009

Trajectory of Fire Balls Noticed East of Tashkent



[redacted]

Part I - Weather Balloons

1. In the analysis of Flyobrpts prior to 1 Jul 52 approximately 15% were classified as "possibly" or "probably" balloon. The basis for decision was generally little more than a form of guesswork; if the Flyobrpt did not do anything, and much leeway was allowed for observer's fallibility, that a balloon could not do in maneuvers, speed, etc., and if the description corresponded even roughly to that of a balloon, it was so classified. If there was no particular reason to believe a balloon was in the area, the report became a "possible". If the sighting occurred near a balloon launching site or on or about the launch time, it became a "probable". It was obvious that an effort to obtain factual data to support such conclusions was in order.

2. ATIAA-5 approached the problem of weather balloons first. Weather balloons are of the following types:

a. Radiosonde - Rubberized tan latex, 6' in diameter at launch, up to 20' at altitude. Carries a transmitter and telemetering device for temperature pressure, dewpoint sequences, which transmitter under certain conditions would give radar returns. Also carries a white running light during night launches battery operated, which should last for duration of flight. Normal ascent is to 70,000' - 100,000', at ± 1,000 ft/min, at which altitude the balloon bursts and equipment recovery is effected by a red parachute.

b. Rawin - Same balloon as above, but it carries only a radar "triangle", and is a winds aloft observation.

c. Rawinsonde - Same, a combination of rawin and radiosonde.

d. Ratal - Same type of balloon, tracked by theodolite for winds aloft observation.

e. Pibal - A rubberized tan latex balloon, 30" in diameter at release and 4 or 5' at altitude. Burst and climb comparable to radiosonde. A winds aloft observation, tracked by theodolite. Carries running light for night launches.

All types of balloons are launched at 0300Z, 0900Z, 1500Z and 2100Z daily. However, some stations launch one, two, three, or four times daily; others launch irregularly, some launch only one type, and others several or all. In addition, time of launch may vary approximately thirty minutes from the scheduled time, either way. All agencies which launch balloons are quick to admit that balloons can malfunction and that many are lost. In addition, wind currents at altitude can cause the balloons to assume odd shapes and strange maneuvers. The balloons under certain atmospheric conditions can appear to be almost any color, and may be visible even at extreme altitudes, particularly at sunrise and sunset, to an observer on the ground.

3. ATIAA-5, faced with this situation, compiled in July a file of balloon launch data cards for Air Weather Service, Naval Air Weather Service, and Weather Bureau launch stations. In addition, this information is pictured graphically on the weather balloon launch location chart. Combining this information with the winds aloft data which ATIC receives from the facsimile charts has often provided a solution to Flyobrpts. Significantly, balloons, possible and probable, increased from 15% in June to 30% in August, with 24% in July. The percentage of reports analyzed as "unknown" decreased proportionately. This gain is a real one, and results from the accumulation of the background data and the elimination of guesswork.

4. The actual radiosonde meteorological information is extracted by all agencies launching balloons onto WBAN 31a, 31b, and 31c. For winds aloft observations, all agencies use WBAN 20 and 20a, and these forms also include the track of the balloon. All agencies forward these records to the National Weather Records Center, Grove Arcade Building, Asheville, North Carolina. ATIAA-5 has requested the CG, AWS, which maintains a detachment at Asheville, to permit "Blue Book" to deal directly with Asheville. The intention is to request photostats of the sounding (WBAN 31a, b, c) and the balloon track (WBAN 20 and 20a) at certain specific times and places. If this is approved, ATIC will be in a position to obtain these records for every balloon flight launched in the U.S., from overseas American bases, and from all the U.S. ships and weather stations at sea. In addition, ATIAA-5 will continue to use the balloon launch information available in this office and will from time to time TWX various launch sites for specific information. These methods of approach will solve the problem of weather balloons.

Part II - Upper Air Research Balloons

1. Specially designed types of balloons are used by the USAF and the U.S. Navy in cooperation with various contractors to obtain upper air data for scientific purposes. There is no doubt that these balloons cause Flyobrpts; tracking data of eleven such flights in July resulted in positive identification in three cases, probable identification in three more. The U.S. Navy, through its field representative of ONR at the University of Minnesota, deals with three contractors. The balloons released are large white polyethylene types capable of expanding to 100' in diameter and carrying up to 500 pounds of metallic equipment. Valve and inflation arrangements control floating altitudes. Naturally, they are visible even at extreme altitudes under many conditions and are capable of assuming almost any shape. The contractors often release from time to time free or attached clusters of the RA and P type rubberized balloons, as well.

2. These flights are often of long duration; one Minneapolis released balloon was tracked to Cape Cod and lost, then it was recovered in Bordeaux, France. They are tracked by ten RDF stations throughout the United States.

3. ATIAA-5 has taken steps to set up a reporting system for all balloon flights of the Navy contractors. This program will be implemented 15 Oct 52 and will permanently solve the problem of U.S. Navy upper air research balloons.

4. The USAF operates two projects, "Gopher" and "Moby Dick", which involve the release of the large polyethylene type balloons. In all particulars, flight durations, tracking methods, etc., those flights are comparable to the U.S. Navy projects. At present, ATIAA-5 has no communication or liaison with these projects, but ATIAA-5 intends to use the same approach and reporting systems with the USAF projects as with the Naval contractors.

Conclusion:

By 1 Nov 52 ATIAA-5 should be receiving complete data on all weather, Navy upper air, and USAF upper air balloon releases.

Note:

This paper is a short introduction to the "balloon phase" of Project Blue Book. For anyone desiring the complete information, such as agencies and personalities involved, channels and methods of communication, etc., it will be necessary to read the following supporting papers which are on file in ATIAA-5.

a. Balloon Data Folder

b. Miscellaneous Correspondence File - Letter 5 Sep 52, to: USAF Cambridge Research Center, Cambridge, Massachusetts, subj: Air Force Upper Air Research Balloon Releases, and first indorsement thereto.

c. Air Weather Service Correspondence File - Letter 22 Sep 52, to: CG, AWS, subj: Climatology Data for Project Blue Book.

d. U.S. Navy Correspondence File - Letter, 9 Sep 52, to: Air Branch, ONR, subj: ONR Upper Air Balloon Projects, and ONR answer thereto.

e. Travel Report - Lt A. G. Flues, 25 Aug 52 to Washington, D.C.

f. Travel Report - Lt A. G. Flues, 15 Sep 52, to Ashevillo, N.C.

g. Travel Report - Lt A. G. Flues, 30 Sep 52, to Minneapolis, Minnesota.